Lesson 20: Observational Research

English Stage 4 (Psychology) · Unit 3: Research Designs

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 6.6: Observational Research; 6.7: Non-Experimental Research (Summary).

In this lesson

Observational research describes behavior by observing and recording it, without manipulating anything. You will learn five observational methods with their strengths, weaknesses and ethical problems, and review Chapter 6 with the book's exercises. In English, you will practice the past passive of method descriptions and the language of definitions.

Learning goals

After this lesson you should be able to:

- Distinguish five observational methods and give a study from the book for each.
- Explain reactivity, the Hawthorne effect and the ethics of disguised observation.
- Describe coding and how researchers show interrater reliability.
- Explain why case studies can inspire theories but cannot support them.
- Describe a procedure in the passive voice and define terms precisely.

Key language

Term or phrase	Meaning	Example
observational research	non-experimental studies in which behavior is systematically observed and recorded to describe variables	the "pace of life" study
naturalistic observation	observing behavior in the environment in which it typically occurs	Goodall's chimpanzees
reactivity	when a measure changes participants' behavior	acting differently when recorded in a bar
Hawthorne effect	reactivity because people know they are being observed and studied	reality-show contestants at first
participant observation	the researcher becomes an active participant in the group or situation studied	Festinger and the Seekers
structured observation	observing a few specific behaviors in a more structured setting, usually to collect quantitative data	Kraut and Johnston's bowlers

Term or phrase	Meaning	Example
coding	specifying target behaviors ahead of time, then watching for and recording them	"closed smile," "look away"
interrater reliability	close agreement between observers who code the same behaviors independently	97% agreement
case study	a detailed description of an individual case	HM
archival data	existing data collected or created for some other purpose	Social Security records
content analysis	analyzing archival data by identifying specific words, phrases or ideas and counting or summarizing them	rated explanations of bad events
as such	for this reason	"nothing is manipulated or controlled, and as such we cannot arrive at causal conclusions"
it is important to note that	introduces a key limit	"it is important to note that ... they should never be used as evidence for theories"

Rules and explanations

The ideas

Observational research gives a "snapshot" of an individual, group or setting. Because nothing is manipulated or controlled, it cannot support causal conclusions. Its data may be qualitative, quantitative or both.

Method	What researchers do	Studies in the book
Naturalistic observation	observe behavior where it typically occurs (field research)	Goodall: three decades with chimpanzees in East Africa
Participant observation	become active members of the group	Festinger et al. (1956); Rosenhan (1973); Wilkins (2008)
Structured observation	observe a few specific behaviors in a lab or structured setting	Levine & Norenzayan (1999); Kraut & Johnston (1979); Cohen et al. (1996)
Case study	examine one individual (or a social unit or event) in depth, often over time	HM; Anna O.; Little Albert
Archival research	analyze existing data	Pelham et al. (2005); Peterson et al. (1988)

Naturalistic observation. *Disguised* observation (participants are unaware) is less reactive, so it can have higher validity. It is ethical if participants remain anonymous and the behavior is public, where people do not expect privacy (grocery shoppers, but not people in a public restroom).

Undisguised observation risks the *Hawthorne effect*, although people often habituate to being observed (think of reality shows).

Participant observation. Some information may be accessible only to an active member of a group. Festinger and colleagues joined the Seekers, a cult that expected the apocalypse on December 21, 1954; when the prophecy failed, members decided that their faith had saved the world (an illustration of cognitive dissonance theory). Rosenhan's pseudopatients were admitted to psychiatric hospitals to observe how staff treated patients. Participant observers gain an insider's view, but they can change the group and become less objective. *Disguised* participant observation involves no informed consent and uses deception, though it may be the only way into a protective group and is less reactive.

Structured observation is efficient and controlled, but less natural settings lower external validity, and observers are often not disguised, which raises concerns about reactivity.

- *Sampling*: when, where, under what conditions and whom to observe. Levine and Norenzayan timed pedestrians over 60 feet (Canada and Sweden: just under 13 seconds; Brazil and Romania: close to 17); observing only on clear summer days controlled for weather.
- *Coding*: target behaviors are defined so that different observers code them the same way. Kraut and Johnston's observers memorized a list of reactions and practiced on videotaped bowlers; two observers agreed 97% of the time. Bowlers rarely smiled facing the pins but often did after turning to their companions, suggesting that smiling is also social communication.
- Cohen et al. (1996): observers rated participants bumped and insulted by a confederate; southerners (from a "culture of honor") reacted with more aggression than northerners.

Case studies give rich detail, are often the only way to study rare conditions, and can give insight into normal behavior. After his 1953 surgery for epilepsy, HM kept his short-term memory but could not consolidate new long-term memories (anterograde amnesia), which suggested that these are different abilities. Still, case studies "should never be used as evidence for theories":

- *internal validity*: no controls, so alternative explanations remain (HM's surgeon may have damaged another area);
- *external validity*: one, usually abnormal, person cannot be generalized;
- *bias*: the researcher's theory can color the description.

Freud's Anna O. is an effective *illustration* of his theory but "essentially worthless" as *evidence*.

Archival research. Pelham et al. used Social Security records to show that women named Virginia, Georgia, Louise and Florence were especially likely to have moved to Virginia, Georgia, Louisiana and Florida. Peterson et al. had raters score explanations of bad events written by male students in the 1940s; the more optimistic men were healthier at about 60 (r = +.25). This is **content analysis**.

Chapter 6 summary. Non-experimental research is either correlational or observational. Internal validity is high in experiments, low in correlational research.

Language focus

1. The past passive in method descriptions. Levine and Norenzayan described their sampling with *was/were* + past participle:

> "Male and female walking speed over a distance of 60 feet *was measured* in at least two locations ... Measurements *were taken* during main business hours on clear summer days. ... only pedestrians walking alone *were used*. Children, individuals with obvious physical handicaps, and window-shoppers *were not timed*."

The passive keeps the focus on the procedure. Use the active when the researcher matters: "Wilkins spent 12 months attending and participating in the group's meetings and social events."

2. Defining a term.

- term + *is* + class + defining feature: "A **case study** is an in-depth examination of an individual."
- *refers to*: "**Reactivity** refers to when a measure changes participants' behavior."
- naming after describing: "This type of reactivity is known as the Hawthorne effect."

> **Note for Arabic speakers** - Arabic nominal sentences need no verb "to be," so learners sometimes drop it from the passive ("Measurements taken during business hours"). A passive sentence needs *be* + past participle: "Measurements *were* taken ..." - Write *refers to* + noun (or *when* + clause), never *refers that*.

Practice

1. Reading questions

- 1.1 When is disguised naturalistic observation considered ethically acceptable?
- 1.2 What is the Hawthorne effect, and why might it fade over time?
- 1.3 What is the only difference between naturalistic and participant observation? What two ethical problems does disguised participant observation raise?
- 1.4 How did Kraut and Johnston show good interrater reliability?
- 1.5 Using HM, explain one internal-validity and one external-validity problem of case studies.
- 1.6 How did Peterson et al. turn the men's questionnaire answers into an explanatory style score?

2. The book's exercises (Section 6.7). Lessons 17 and 18 contain three others. For 2.4, you may use Google Scholar instead of PsycINFO.

- 2.1 Discussion: For each of the following studies, decide which type of research design it is and explain why. (a) A researcher conducts detailed interviews with unmarried teenage fathers to learn about how they feel and what they think about their role as fathers and summarizes their feelings in a written narrative. (b) A researcher measures the impulsivity of a large sample of drivers and looks at the statistical relationship between this variable and the number of traffic tickets the drivers have received. (c) A researcher randomly assigns patients with low back pain either to a treatment involving hypnosis or to a treatment involving exercise. She then measures their level of low back pain after 3 months.
- 2.2 Discussion: For each of the following, decide whether it is most likely that the study described is experimental or non-experimental and explain why. (The five studies are in Section 6.7: word

recall, grade point averages, stick shifts, refrigerators, helping.)

- 2.3 Discussion: What are some ways in which a qualitative study of girls who play youth baseball would likely differ from a quantitative study on the same topic? How would the data differ by interviewing girls one-on-one rather than conducting focus groups or surveys?
- 2.4 Practice: Find and read a published case study in psychology. (Use *case study* as a key term in a PsycINFO search.) Then (a) describe one problem related to internal validity, (b) describe one problem related to external validity, and (c) generate one hypothesis suggested by the case study that might be interesting to test in a subsequent study.

3. Language: rewrite in the past passive (leave out the agent unless it matters). (a) The researchers identified all negative events in the questionnaire responses. (b) A separate group of raters rated each explanation on three dimensions. (c) They measured out a 60-foot distance along a city sidewalk. (d) Kraut and Johnston video recorded a subset of their participants' reactions.

Speak with your tutor

- Ask the tutor to describe five short studies; name the observational method in each and justify your answer.
- Discuss the ethics of Festinger's and Rosenhan's disguised participant observation, using Lessons 07-09. Could they be approved today?
- In PSYC 110 Lecture 11 (Emotions, Part I), Professor Bloom describes filmed bowlers who grin only when they turn to their friends. Compare this with Kraut and Johnston's method: why was coding needed?
- In PSYC 110 Lecture 12 (Emotions, Part II), Bloom describes the culture-of-honor hallway study and reports stress hormones (cortisol, testosterone); the book reports observers' ratings. Why do several measures of one reaction strengthen a study?
- In PSYC 110 Lecture 16 (Self and Other, Part I), Bloom presents Festinger's experiment in which people paid $1 rated a boring task as more fun than people paid $20. Why is this stronger evidence for cognitive dissonance theory than the Seekers study?

Writing task

Research project: an observational alternative (180-250 words). Describe, as a short Method paragraph, one observational method that could study sleep and memory in university students: for example, structured observation (coding signs of sleepiness in early-morning lectures), archival research (existing records) or a case study (a student with a rare sleep disorder). Include:

1. the method and why it fits your question;
2. sampling: when, where, under what conditions and whom you will observe;
3. at least three target behaviors, each defined in one sentence (or the content you will look for in archival data);
4. how two observers will code independently and report their agreement;
5. ethics and reactivity: disguised or not, public or private setting, anonymity;

6. one limitation, including why no causal conclusion is possible.

Use the future passive (*will be observed*, *will be coded*) and a definition pattern for each behavior. If you work on your own topic, write the same parts for it.

Self-editing checklist

- Is the method named correctly and suited to the question?
- Did you say when, where, under what conditions and whom?
- Could two observers code each behavior the same way, and will you check their agreement?
- Did you address privacy and reactivity?
- Did you avoid causal words (*causes*, *leads to*)?
- Does every passive have *be* + past participle?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- What happens when you remove the hippocampus? (TED-Ed, Sam Kean), on HM: https://youtu.be/KkaXNvzE4pk
- The New York Times (2016) article on the controversies after HM's death (archived copy): https://web.archive.org/web/20251003023644/https://www.nytimes.com/2016/08/07/magazine/the-brain-that-couldnt-remember.html

Answer key

1. Reading questions

- 1.1 When participants remain anonymous and the behavior occurs in a public setting where people would not normally expect privacy ("Naturalistic Observation").
- 1.2 Reactivity that occurs because people know they are being observed and studied. Over time, people habituate to being observed and behave naturally.
- 1.3 Participant observers become active members of the group. Disguised participant observation obtains no informed consent and uses deception.
- 1.4 Two observers independently coded a video-recorded subset of reactions and agreed 97% of the time.
- 1.5 Internal: without controls, alternatives cannot be ruled out (the surgeon may have damaged another brain area). External: HM was one abnormal individual, so the results cannot be generalized.
- 1.6 Negative events and their causal explanations were written on index cards, separate raters rated each explanation on three optimism-pessimism dimensions, and the ratings were averaged.

2. The book's exercises

- 2.1 (a) Qualitative research: detailed interviews about feelings and thoughts, summarized in a written narrative rather than with statistics. (b) Correlational research: two variables are measured and their statistical relationship examined; nothing is manipulated. (c) Experimental research: the treatment (hypnosis vs. exercise) is manipulated and patients are randomly assigned to it.
- 2.2 Experimental (an IV is manipulated): word recall (read vs. imagine), stick shifts, helping (hurry vs. take your time). Non-experimental (both variables are measured): grade point averages and performance; refrigerator temperature and bacteria.
- 2.3 (example answer) A qualitative study would ask a broad question (what it is like to be a girl on a baseball team), interview a few girls at length and report themes with quotations. A quantitative study would test a focused question (for example, whether playing is related to self-esteem) with numerical measures from a large sample. One-on-one interviews give each girl's experience in depth, free from peer influence; focus groups may bring out more through interaction, but the wish to be liked and dominant, extraverted girls can distort answers; surveys give comparable answers from many girls but little depth.
- 2.4 (example answer, using HM from Section 6.6; yours depends on the case you read) (a) The surgeon may have damaged another area, so the hippocampus may not be the only cause of the amnesia. (b) HM was one person with severe epilepsy and unusual surgery, so his case may not generalize. (c) Compared with people whose brain damage lies elsewhere, people with damage limited to the hippocampus will have more difficulty forming new long-term memories but normal short-term memory.

3. Language (a) All negative events in the questionnaire responses were identified. (b) Each explanation was rated on three dimensions by a separate group of raters (keep the agent: it matters

that the raters were separate). (c) A 60-foot distance was measured out along a city sidewalk. (d) A subset of the participants' reactions was video recorded.

My notes